Linear Algebra and Its Applications

Section 3.1: Introduction to determinants

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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3.1 Introduction to determinants

A determinant is a number attached to a square matrix. For a 2×2 matrix it is $ad - bc$. In a larger matrix, delete row i and column j to obtain the minor matrix A_{ij}; its cofactor is the *number* $C_{ij} = (-1)^{i+j} \det A_{ij}$. The signs form a checkerboard, beginning with plus in position $(1, 1)$.

A cofactor expansion multiplies every entry of one chosen row or column by its cofactor, then adds. A zero entry contributes zero. Row or column positions in each new minor must be counted anew: they are not the original positions. For numerical checks below all arithmetic was also performed exactly with rational-number matrix computations.

Practice problem 3.1.P1: A sparse fourth-order determinant

What we must show. Compute the determinant without expanding sixteen terms.

Plan. Column 3 has just one nonzero entry, 2 in row 1. Its sign is $(-1)^{1+3} = 1$.

$$D = 2 \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}.$$

Expand that minor down its first column. The only nonzero entry is now -5 in position $(2, 1)$, whose cofactor sign is negative. Hence

$$D = 2[-(-5)] \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} = 10[3(-6) - (-4)5] = 10(2) = 20.$$

Conclusion. The determinant is 20. Keeping track of the sign and the negative entry separately prevents a common sign error.

Exercise 3.1.1: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (3)(-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 5 & -1 \end{vmatrix} + (4)(-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 0 & 5 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{11} = (3)(-1) - (2)(5) = -13$$
$$\det A_{13} = (2)(5) - (3)(0) = 10.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{2+2}\begin{vmatrix}3 & 4\\ 0 & -1\end{vmatrix} + (5)(-1)^{3+2}\begin{vmatrix}3 & 4\\ 2 & 2\end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{22} = (3)(-1) - (4)(0) = -3$$
$$\det A_{32} = (3)(2) - (4)(2) = -2.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 1$.

Exercise 3.1.2: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix}0 & 5 & 1\\ 4 & -3 & 0\\ 2 & 4 & 1\end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix}4 & 0\\ 2 & 1\end{vmatrix} + (1)(-1)^{1+3}\begin{vmatrix}4 & -3\\ 2 & 4\end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{13} = (4)(4) - (-3)(2) = 22.$$

Check by expanding down column 2, as requested.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix}4 & 0\\ 2 & 1\end{vmatrix} + (-3)(-1)^{2+2}\begin{vmatrix}0 & 1\\ 2 & 1\end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix}0 & 1\\ 4 & 0\end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{22} = (0)(1) - (1)(2) = -2$$
$$\det A_{32} = (0)(0) - (1)(4) = -4.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 2$.

Exercise 3.1.3: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix} + (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (3)(-1)^{1+3}\begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (1)(-1) - (2)(4) = -9 \\ \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{13} &= (3)(4) - (1)(1) = 11. \end{aligned}$$

Check by expanding down column 2, as requested.

$$\det A = (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{22} &= (2)(-1) - (3)(1) = -5 \\ \det A_{32} &= (2)(2) - (3)(3) = -5. \end{aligned}$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = -5$.

Exercise 3.1.4: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (5)(-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{11} = (1)(2) - (1)(4) = -2$$
$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{13} = (2)(4) - (1)(3) = 5.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 1 & 5 \\ 3 & 2 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{22} = (1)(2) - (5)(3) = -13$$
$$\det A_{32} = (1)(1) - (5)(2) = -9.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 20$.

Exercise 3.1.5: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 4 & 0 & 5 \\ 5 & 1 & 6 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} + (-4)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 5 & 1 \end{vmatrix} = -23.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(6) - (5)(1) = -5$$
$$\det A_{12} = (4)(6) - (5)(5) = -1$$
$$\det A_{13} = (4)(1) - (0)(5) = 4.$$

Conclusion. $\det A = -23$.

Exercise 3.1.6: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 5 & -2 & 4 \\ 0 & 3 & -5 \\ 2 & -4 & 7 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+1}\begin{vmatrix} 3 & -5 \\ -4 & 7 \end{vmatrix} + (-2)(-1)^{1+2}\begin{vmatrix} 0 & -5 \\ 2 & 7 \end{vmatrix} + (4)(-1)^{1+3}\begin{vmatrix} 0 & 3 \\ 2 & -4 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (3)(7) - (-5)(-4) = 1 \\ \det A_{12} &= (0)(7) - (-5)(2) = 10 \\ \det A_{13} &= (0)(-4) - (3)(2) = -6. \end{aligned}$$

Conclusion. $\det A = 1$.

Exercise 3.1.7: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 3 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (4)(-1)^{1+1}\begin{vmatrix} 5 & 2 \\ 7 & 3 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 6 & 2 \\ 9 & 3 \end{vmatrix} = 4.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (5)(3) - (2)(7) = 1 \\ \det A_{12} &= (6)(3) - (2)(9) = 0. \end{aligned}$$

Conclusion. $\det A = 4$.

Exercise 3.1.8: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 8 & 1 & 6 \\ 4 & 0 & 3 \\ 3 & -2 & 5 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (8)(-1)^{1+1}\begin{vmatrix} 0 & 3 \\ -2 & 5 \end{vmatrix} + (1)(-1)^{1+2}\begin{vmatrix} 4 & 3 \\ 3 & 5 \end{vmatrix} + (6)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 3 & -2 \end{vmatrix} = -11.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(5) - (3)(-2) = 6$$
$$\det A_{12} = (4)(5) - (3)(3) = 11$$
$$\det A_{13} = (4)(-2) - (0)(3) = -8.$$

Conclusion. $\det A = -11$.

Exercise 3.1.9: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix} = (2)(-1)^{3+1} D_2 = 10.$$

$$D_2 = \begin{vmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{vmatrix} = (5)(-1)^{1+3} D_3 = 5.$$

$$D_3 = \begin{vmatrix} 7 & 2 \\ 3 & 1 \end{vmatrix} = (7)(1) - (2)(3) = 1.$$

Thus the answer is 10.

Exercise 3.1.10: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix} = (3)(-1)^{2+3} D_2 = -6.$$

$$D_2 = \begin{vmatrix} 1 & -2 & 2 \\ 2 & -6 & 5 \\ 5 & 0 & 4 \end{vmatrix} = (5)(-1)^{3+1} D_3 + (4)(-1)^{3+3} D_4 = 2.$$

$$D_4 = \begin{vmatrix} 1 & -2 \\ 2 & -6 \end{vmatrix} = (1)(-6) - (-2)(2) = -2.$$

$$D_3 = \begin{vmatrix} -2 & 2 \\ -6 & 5 \end{vmatrix} = (-2)(5) - (2)(-6) = 2.$$

Thus the answer is -6.

Exercise 3.1.11: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{vmatrix} = (2)(-1)^{4+4} D_2 = -12.$$

$$D_2 = \begin{vmatrix} 3 & 5 & -8 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{vmatrix} = (1)(-1)^{3+3} D_3 = -6.$$

$$D_3 = \begin{vmatrix} 3 & 5 \\ 0 & -2 \end{vmatrix} = (3)(-2) - (5)(0) = -6.$$

Thus the answer is -12.

Exercise 3.1.12: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the

deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix} = (4)(-1)^{1+1} D_2 = 36.$$

$$D_2 = \begin{vmatrix} -1 & 0 & 0 \\ 6 & 3 & 0 \\ -8 & 4 & -3 \end{vmatrix} = (-1)(-1)^{1+1} D_3 = 9.$$

$$D_3 = \begin{vmatrix} 3 & 0 \\ 4 & -3 \end{vmatrix} = (3)(-3) - (0)(4) = -9.$$

Thus the answer is 36.

Exercise 3.1.13: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = (2)(-1)^{2+3} D_2 = 6.$$

$$D_2 = \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & -8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix} = (3)(-1)^{2+2} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} = (-1)(-1)^{3+2} D_4 + (2)(-1)^{3+3} D_5 = -1.$$

$$D_5 = \begin{vmatrix} 4 & 3 \\ 5 & 2 \end{vmatrix} = (4)(2) - (3)(5) = -7.$$

$$D_4 = \begin{vmatrix} 4 & -5 \\ 5 & -3 \end{vmatrix} = (4)(-3) - (-5)(5) = 13.$$

Thus the answer is 6.

Exercise 3.1.14: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix} = (3)(-1)^{4+1} D_2 = 9.$$

$$D_2 = \begin{vmatrix} 3 & 2 & 4 & 0 \\ 0 & -4 & 1 & 0 \\ -5 & 6 & 7 & 1 \\ 2 & 3 & 2 & 0 \end{vmatrix} = (1)(-1)^{3+4} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 3 & 2 & 4 \\ 0 & -4 & 1 \\ 2 & 3 & 2 \end{vmatrix} = (-4)(-1)^{2+2} D_4 + (1)(-1)^{2+3} D_5 = 3.$$

$$D_5 = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = (3)(3) - (2)(2) = 5.$$

$$D_4 = \begin{vmatrix} 3 & 4 \\ 2 & 2 \end{vmatrix} = (3)(2) - (4)(2) = -2.$$

Thus the answer is 9.

Exercise 3.1.15: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}$. The three downward products are -9, 0, 40, and the three upward products are 0, 30, 0. Add the downward products and subtract the entire upward sum:

$$\det A = (-9 + 0 + 40) - (0 + 30 + 0) = 1.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.16: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix}$. The three downward products are 0, 0, 16, and the three upward products are -6, 0, 20. Add the downward products and subtract the entire upward sum:

$$\det A = (0 + 0 + 16) - (-6 + 0 + 20) = 2.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.17: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}$. The three downward products are -2, -8, 36, and the three upward products are 3, 16, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (-2 + -8 + 36) - (3 + 16 + 12) = -5.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.18: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}$. The three downward products are 2, 9, 40, and the three upward products are 15, 4, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (2 + 9 + 40) - (15 + 4 + 12) = 20.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.19: A row operation and its determinant

The second matrix is obtained by $R_1 \leftrightarrow R_2$. Its determinant is $cb - da = -(ad - bc)$. Thus a row interchange reverses the determinant's sign.

Exercise 3.1.20: A row operation and its determinant

The operation is $R_2 \leftarrow kR_2$. Direct calculation gives $a(kd) - b(kc) = k(ad - bc)$. Multiplying one row by k multiplies the determinant by k.

Exercise 3.1.21: A row operation and its determinant

The operation is $R_2 \leftarrow R_2 + kR_1$. The old determinant is $3 \cdot 6 - 4 \cdot 5 = -2$; the new one is $3(6 + 4k) - 4(5 + 3k) = 18 + 12k - 20 - 12k = -2$. The added terms cancel, so the determinant is unchanged.

Exercise 3.1.22: A row operation and its determinant

The operation is $R_1 \leftarrow R_1 + kR_2$. The new determinant is $(a + kc)d - (b + kd)c = ad + kcd - bc - kdc = ad - bc$. Thus row replacement preserves the determinant.

Exercise 3.1.23: A row operation and its determinant

The first row $(1, 1, 1)$ is replaced by $(k, k, k) = k(1, 1, 1)$. To see the factor directly, expand along row 1: every term in the expansion gains exactly one factor k. Thus the new determinant is k times the old one (also when $k = 0$).

Exercise 3.1.24: A row operation and its determinant

The operation interchanges rows 1 and 2. Interchanging these two rows reverses the sign of every signed product in the determinant, so the new determinant is the negative of the old determinant.

Exercise 3.1.25: An elementary-matrix determinant

This lower triangular matrix has diagonal $1, 1, 1$. Its determinant is their product, 1. It performs $R_3 \leftarrow R_3 + kR_2$ on another matrix.

Exercise 3.1.26: An elementary-matrix determinant

The matrix is lower triangular with diagonal $1, 1, 1$, so its determinant is 1. Its row operation is $R_3 \leftarrow R_3 + kR_1$.

Exercise 3.1.27: An elementary-matrix determinant

The diagonal matrix has diagonal $k, 1, 1$, so its determinant is k. It scales row 1 by k. For an elementary matrix this operation requires $k \neq 0$; the displayed determinant formula is still valid at $k = 0$.

Exercise 3.1.28: An elementary-matrix determinant

The diagonal entries are $1, k, 1$, giving determinant k. The operation scales row 2 by k; an elementary scaling has $k \neq 0$.

Exercise 3.1.29: An elementary-matrix determinant

The matrix is obtained from I_3 by interchanging rows 1 and 2, so its determinant is -1. Direct expansion along row 3 gives $1 \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.30: An elementary-matrix determinant

The matrix interchanges rows 1 and 3 of I_3. Its determinant is -1. Expansion across row 2 gives $1(-1)^{2+2} \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.31: An elementary-matrix determinant

A row-replacement elementary matrix is obtained from I_n by inserting a scalar k into an off-diagonal position (i, j). Expand by rows different from i, j to reduce to the 2×2 block $\left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$ or its transpose. The determinant is 1. Equivalently, adding k times one identity row to another creates no new nonzero signed product. The examples in Exercises 25 and 26 illustrate this general result.

Exercise 3.1.32: An elementary-matrix determinant

A scaling elementary matrix is diagonal with one diagonal entry $k \neq 0$ and all others 1. The triangular-matrix theorem therefore gives determinant k. It is one row that is scaled, not all n rows.

Exercise 3.1.33: Verify multiplication for one elementary matrix

Here $E = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, so $EA = \left[\begin{smallmatrix} c & d \\ a & b \end{smallmatrix}\right]$. Consequently $\det(EA) = cb - da = -(ad - bc)$, while $\det E = -1$. Thus $\det(EA) = (\det E)(\det A)$.

Exercise 3.1.34: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ 0 & k \end{smallmatrix}\right]$, we have $EA = \left[\begin{smallmatrix} a & b \\ kc & kd \end{smallmatrix}\right]$. Its determinant is $akd - bkc = k(ad - bc)$. Since $\det E = k$, the required equality follows.

Exercise 3.1.35: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a+kc & b+kd \\ c & d \end{smallmatrix}\right]$. Its determinant is $ad+kcd-bc-kdc = ad-bc$. Also $\det E = 1$, so the two sides agree.

Exercise 3.1.36: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ k & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a & b \\ ka+c & kb+d \end{smallmatrix}\right]$. Its determinant is $a(kb+d)-b(ka+c) = ad-bc$. This equals $(\det E)(\det A)$ because $\det E = 1$.

Exercise 3.1.37: Determinants and their interpretation

Scalar multiplication scales every entry, so $5A = \left[\begin{smallmatrix} 15 & 5 \\ 20 & 10 \end{smallmatrix}\right]$. We find $\det A = 3 \cdot 2 - 1 \cdot 4 = 2$, but $\det(5A) = 15 \cdot 10 - 5 \cdot 20 = 50$, whereas $5 \det A = 10$. Thus the proposed equality is false. Both rows were multiplied by 5, giving a factor 5^2.

Exercise 3.1.38: Determinants and their interpretation

Write $kA = \left[\begin{smallmatrix} ka & kb \\ kc & kd \end{smallmatrix}\right]$. Then $\det(kA) = (ka)(kd) - (kb)(kc) = k^2(ad - bc) = k^2 \det A$. This calculation also covers $k = 0$.

Exercise 3.1.39: Determinants and their interpretation

(a) True, with the usual initial case.

For $n \geq 2$, the recursive definition expands into determinants of $(n-1) \times (n-1)$ minor matrices. A 1×1 determinant is its sole entry, the base case.

(b) False: a cofactor is a scalar.

Deleting a row and column produces the matrix A_{ij}. The cofactor is $(-1)^{i+j} \det A_{ij}$, a signed number, not that submatrix.

Exercise 3.1.40: Determinants and their interpretation

(a) False.

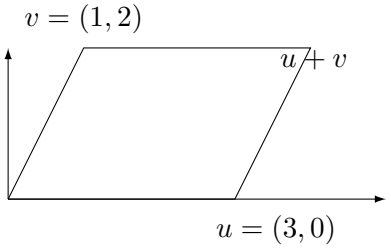
Row and column expansions compute the same number $\det A$, not negatives in general. For example, both expansions of I_2 give 1.

(b) False.

For a triangular matrix the determinant is the *product* of diagonal entries. For I_2 the determinant is 1, whereas the sum of the diagonal is 2.

Exercise 3.1.41: Determinants and their interpretation

What we must show. Compare unsigned area with a determinant, and then vary the shear. The two adjacent edges are $u = (3, 0)$ and $v = (1, 2)$. The horizontal base has length 3 and perpendicular height 2, so the area is $3 \cdot 2 = 6$. Also $\det[u\ v] = 3 \cdot 2 - 0 \cdot 1 = 6$. Replacing v by $(x, 2)$ leaves its height above the horizontal base equal to 2, hence leaves the area 6; the determinant is still $3 \cdot 2 - 0x = 6$.

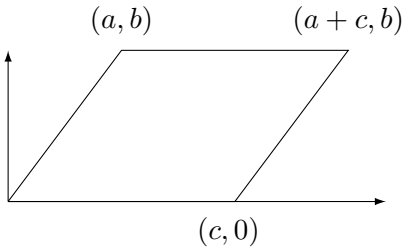


The horizontal displacement changes the slant, not the height. This operation is a shear; the sketch uses $x = 1$, and the argument works for every real x.

Exercise 3.1.42: Determinants and their interpretation

Take the horizontal base $v = (c, 0)$ of length $c > 0$. The other edge $u = (a, b)$ has height $b > 0$, so the area is bc. On the other hand,

$$\det[u\ v] = \begin{vmatrix} a & c \\ b & 0 \end{vmatrix} = -bc, \qquad \det[v\ u] = \begin{vmatrix} c & a \\ 0 & b \end{vmatrix} = bc.$$



The sketch is schematic. Interchanging the order of the columns changes orientation and the determinant's sign, but the ordinary area is nonnegative and equals either determinant's absolute value.

Exercise 3.1.43: A reproducible test of determinant additivity

What we must show. Test additivity with four reproducible randomly generated pairs. A computer experiment supplies evidence, not a proof of a universal identity. Here is an exact integer experiment; a fixed seed lets another reader repeat the same draws.

```
from random import Random
from sympy import Matrix
r = Random(31043)
for n in (5,2,3,4):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    B = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    print(n, A.det(), B.det(), (A+B).det(),
          (A+B).det()-A.det()-B.det())
```

n	$\det A$	$\det B$	$\det(A+B)$	difference
5	10070	136248	121574	−24744
2	−34	1	−78	−45
3	888	−55	1464	631
4	176	748	−6819	−7743

The nonzero differences disprove additivity. A particularly short exact counterexample is $A = B = I_2$: $\det(A + B) = 4$, but $\det A + \det B = 2$. The determinant is linear in *one row at a time with the other rows fixed*, not linear in the entire matrix.

Exercise 3.1.44: A reproducible test of multiplication

Use the same four random pairs and seed from Exercise 43. Replace the last line of that code by

```
print(n, (A*B).det(), A.det()*B.det(),
      (A*B).det()-A.det()*B.det())
```

The exact outputs are

n	$\det(AB)$	$(\det A)(\det B)$	difference
5	1372017360	1372017360	0
2	−34	−34	0
3	−48840	−48840	0
4	131648	131648	0

All four differences are zero. This suggests $\det(AB) = (\det A)(\det B)$. The next section proves it for all square matrices, so the experiment is not being substituted for a proof.

Exercise 3.1.45: Scaling and transposition in several dimensions

Generate three 4×4 samples and two each of sizes 5 and 6. The following code carries out all comparisons, including transpose, with exact arithmetic.

```
from random import Random
from sympy import Matrix
r = Random(31045)
for n in (4,4,4,5,5,6,6):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    d = A.det()
    assert A.T.det() == d
    print(n,d,(-A).det(),(2*A).det(),(10*A).det())
```

The transpose check passed for every sample; the remaining outputs are

n	$\det A$	$\det(-A)$	$\det(2A)$	$\det(10A)$
4	2446	2446	39136	24460000
4	3572	3572	57152	35720000
4	2144	2144	34304	21440000
5	150900	−150900	4828800	15090000000
5	29441	−29441	942112	2944100000
6	−122640	−122640	−7848960	−122640000000
6	−191568	−191568	−12260352	−191568000000

The conjectures are $\det A^T = \det A$ and $\det(kA) = k^n \det A$. In particular, negation preserves the determinant in even dimensions and reverses its sign in odd dimensions. Scaling the matrix scales *every* row, so the scale factor occurs n times. The integer calculations above check that description at every sampled size.

Exercise 3.1.46: A determinant of an inverse

Use seed 31046 and the same integer generator as Exercise 45, now at sizes $4, 5, 6$. For each generated matrix, compute

```
d = A.det()
if d != 0:
    print(n,d,A.inv().det())
else:
    print(A.echelon_form())
```

The three sampled matrices were nonsingular and gave

n	$\det A$	$\det(A^{-1})$
4	-1586	$-\frac{1}{1586}$
5	34641	$\frac{1}{34641}$
6	-48084	$-\frac{1}{48084}$

Each last entry is the reciprocal of the preceding one. Thus the conjecture is $\det(A^{-1}) = 1/\det A$ when A is invertible. A zero determinant would instead lead to a missing pivot in echelon form and no inverse; the reciprocal expression would then be undefined. Multiplicativity in the next section proves the conjecture by applying it to $AA^{-1} = I$.